Analytics & Big Data

Session 1: The rise of analytics

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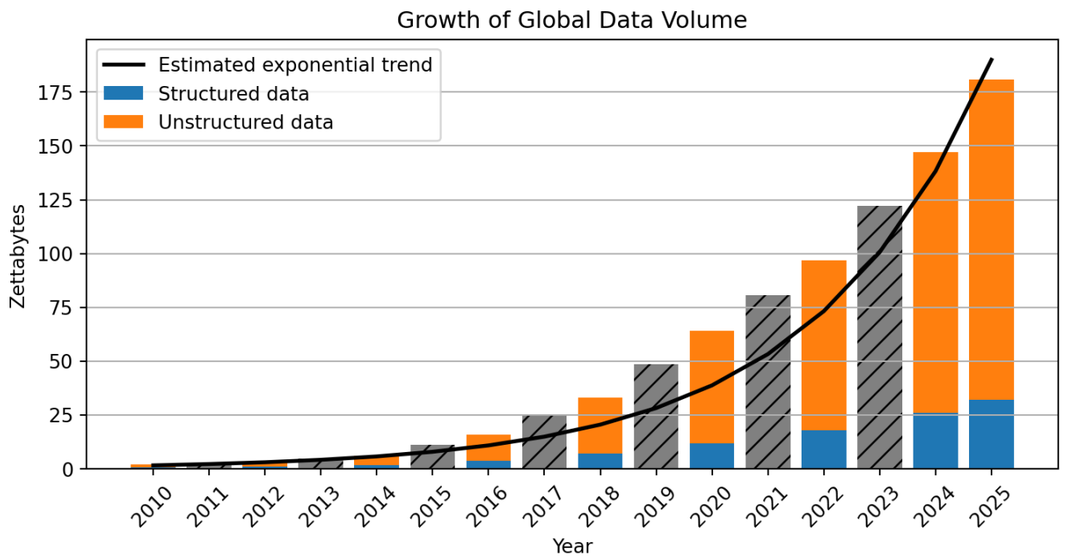
(2026-03-23)

Why is Modern Analytics so Successful?

1. More Data
2. More Computing Power
3. New Algorithms
4. New Analytics Processes

More Data for the Analysis

The Data Explosion



Unit	Equivalent	Approximate Meaning
Gigabyte (GB)		A HD movie file or a few hundred photos
Terabyte (TB)	1,000 GB	Storage of a modern laptop or external drive
Petabyte (PB)	1,000 TB	Data of a large company or several large data centers
Exabyte (EB)	1,000 PB	Roughly the yearly internet traffic of a small country
Zettabyte (ZB)	1,000 EB	≈ 1 trillion gigabytes; global data creation scale

Data production

Enterprise and Transactional Data
Enterprise systems such as ERP, CRM, and supply chain platforms generate large volumes of structured data through everyday business transactions.

eCommerce
Every search, click, purchase, and review creates behavioral and transactional data used for recommendations and personalized marketing.

Social Media and User-Generated Content
Platforms such as TikTok, YouTube, and Instagram generate enormous data volumes through uploads, interactions, and live streaming.

IoT and Smart Devices
Connected devices—from wearables to industrial sensors—continuously produce real-time data across interconnected systems.

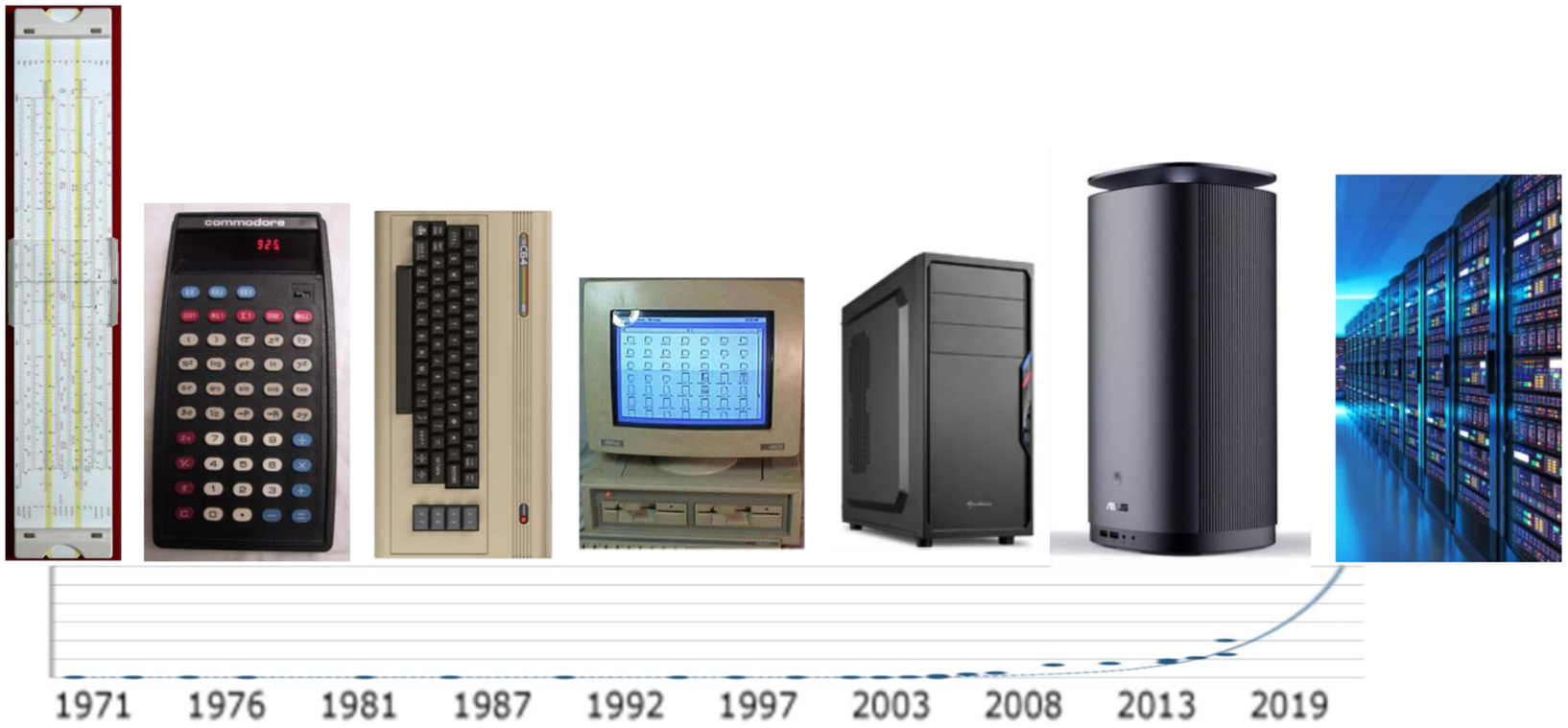
Digital Transactions
Online banking, mobile payments, and blockchain systems generate detailed financial records for transactions and security monitoring.

AI-Generated Data
Machine learning and generative AI create large datasets during training and operation, further accelerating global data growth.

Computing Power

Growth of Computing Power

The rapid acceleration of computing power—driven by advances in hardware, cloud infrastructure, and parallel processing—has enabled modern analytics and machine learning to scale to massive datasets.



Evolution of Computing Power

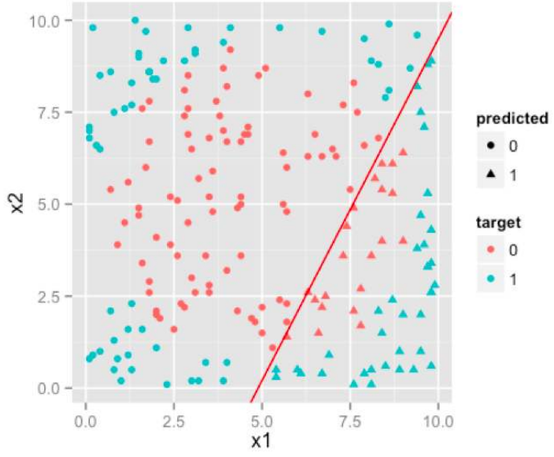
The growth of modern analytics is enabled by **changing strategies for increasing computing power**.

Era	Main Strategy	Explanation
1970s–2000s (*)	**Moore's Law & miniaturization**	Smaller transistors → more components per chip → faster processors
2005–today	**Parallel computing**	Performance increases by using multiple processors simultaneously
2010s–today	**Specialized hardware**	Chips optimized for specific workloads (GPUs, TPUs, AI accelerators)
Emerging	**New computing paradigms**	Alternative computing models such as quantum computing

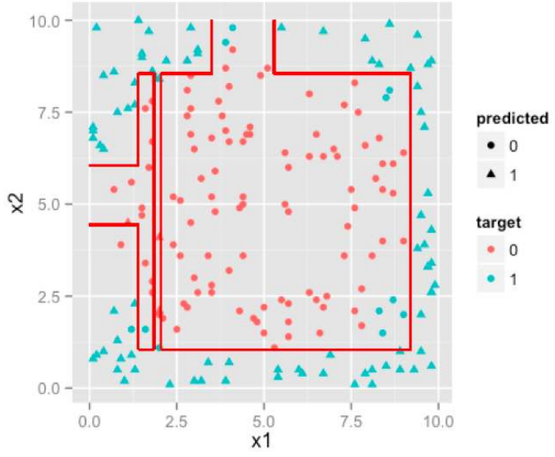
(*) Miniaturization continues today, but progress has slowed as transistors approach physical limits. Advances such as extreme ultraviolet (EUV) lithography are required to continue scaling. See how $400 million machines are constructed to build modern chips: https://www.youtube.com/watch?v=MiUHjLxm3V0.

New Algorithms

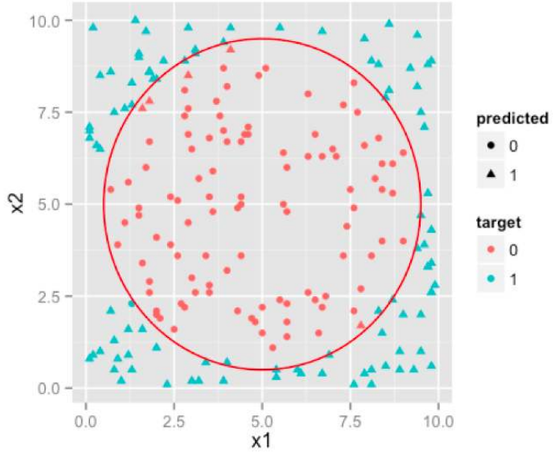
Algorithms enable more powerful models



Traditional Regression



Decision Tree



Neural Network

Rapid Advances in Algorithms

Recent breakthroughs in AI show how new algorithms can rapidly surpass human performance in specific domains.

AlphaGo (2016)

- Developed by **DeepMind**
- Uses **deep neural networks + reinforcement learning**
- Defeated world champion **Lee Sedol** in the game of Go

Key insight:

- Go was long considered too complex for computers due to the enormous search space.

Result

> AI surpassed human experts in complex strategic reasoning.

AlphaFold (2020–2022)

- Developed by **DeepMind**
- Uses deep learning to predict **protein structures**
- Achieved breakthrough performance in the **CASP competition**

Key insight:

- Predicting protein folding had been a major unsolved problem in biology for decades.

Result

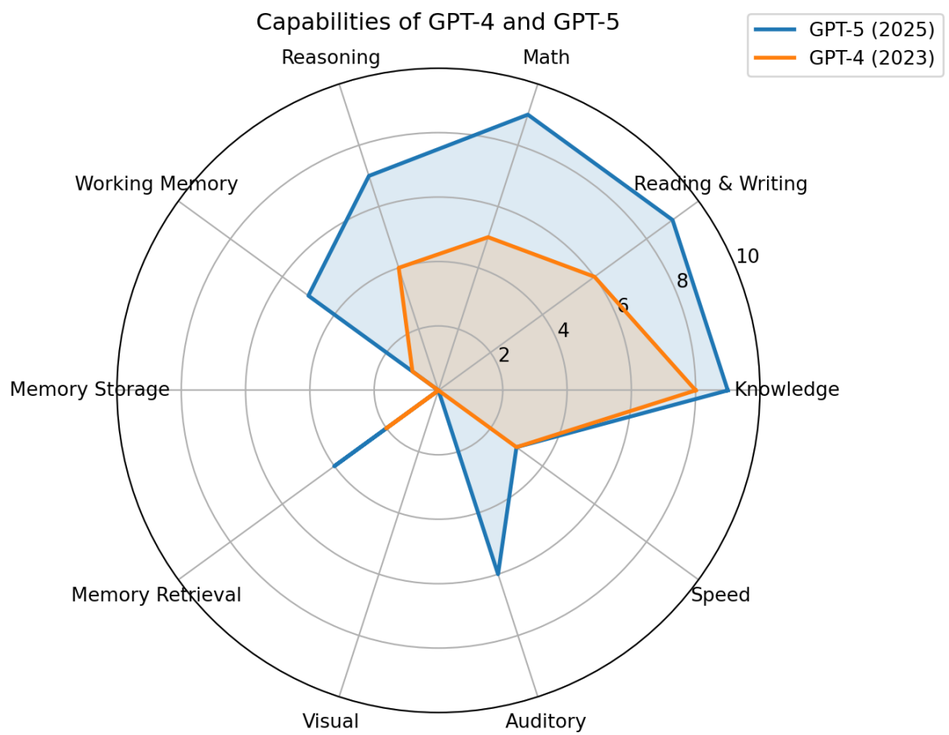
> AI can solve scientific problems that previously required years of human research.

Key takeaway

AI progress often occurs through **algorithmic breakthroughs**, enabling machines to outperform humans in increasingly complex domains:

TODO: - add illustrations - - nobel prize + biotech. ?

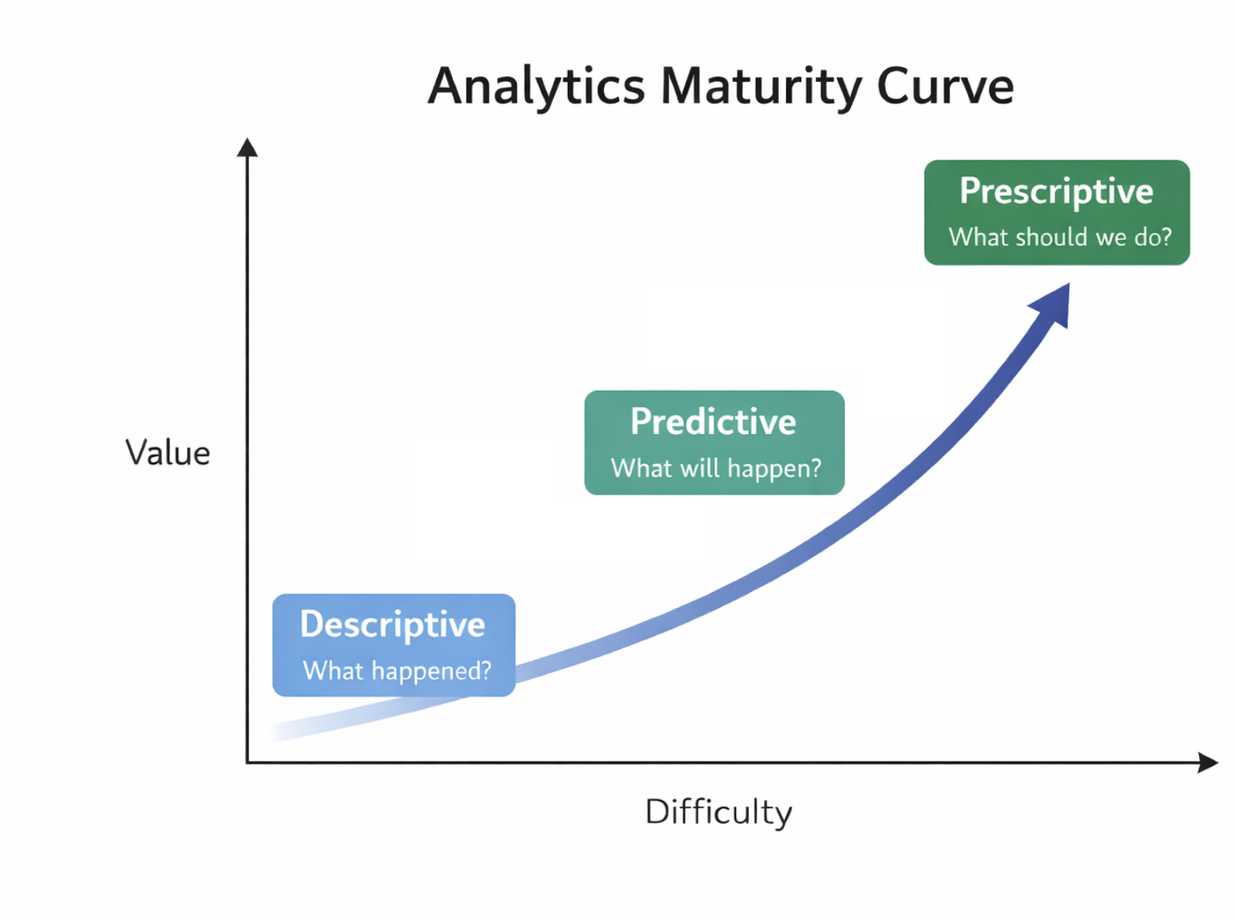
Jagged frontier



TODO: cite Dellacqua

Analytics Processes

Maturing analytical capabilities



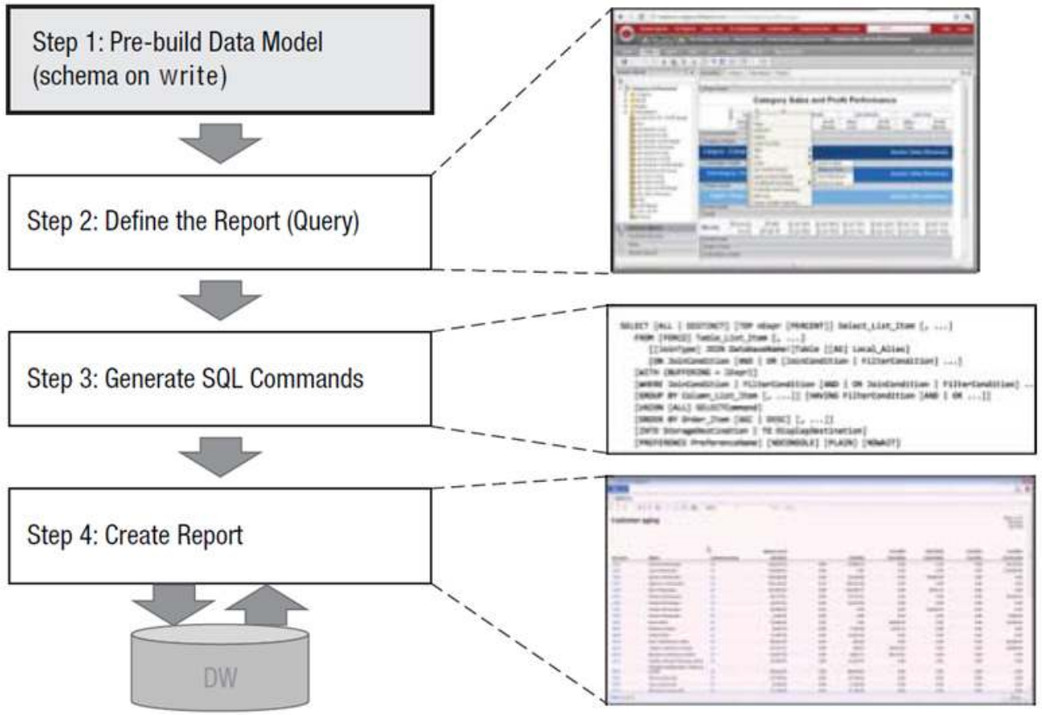
Analytical purposes

What happened? (descriptive)	What will happen? (predictive analytics)	What should I do? (prescriptive analytics)
How many widgets did I sell last month?	How many widgets will I sell next month?	Order **5,000** units of Component Z to support widget sales for next month.
What were sales by zip code for Christmas last year?	What will be sales by zip code over this Christmas season?	Hire **Y** new sales reps by these zip codes to handle projected Christmas sales.
How many of Product X were returned last month?	How many of Product X will be returned next month?	Set aside **$125K** in financial reserve to cover Product X returns.
What were company revenues and profits for the past quarter?	What are projected company revenues and profits for next quarter?	Sell the following product mix to achieve quarterly revenue and margin goals.
How many employees did I hire last year?	How many employees will I need to hire next year?	Increase hiring pipeline by **35%** to achieve hiring goals.

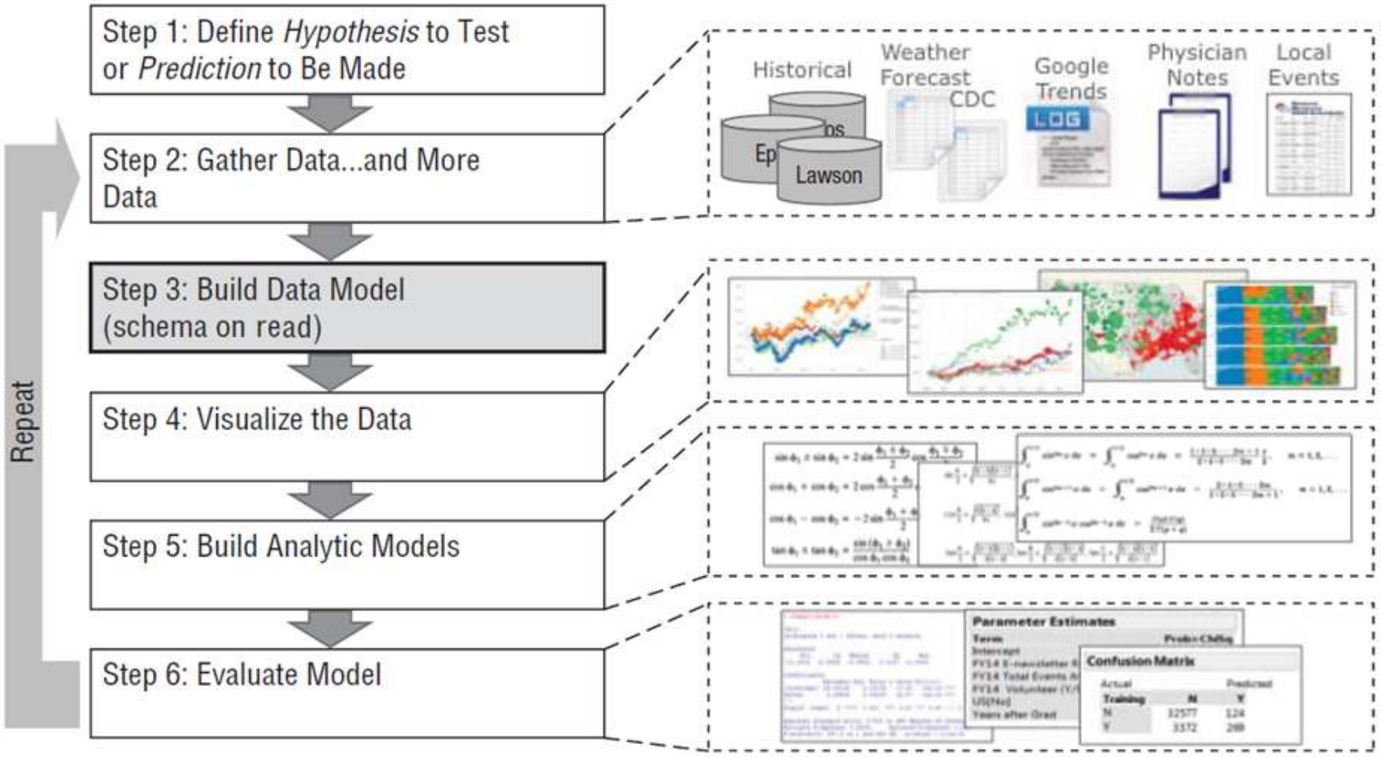
A particular method, such as regression or machine learning, can serve multiple purposes.

Source: Schmarzo, p. 13

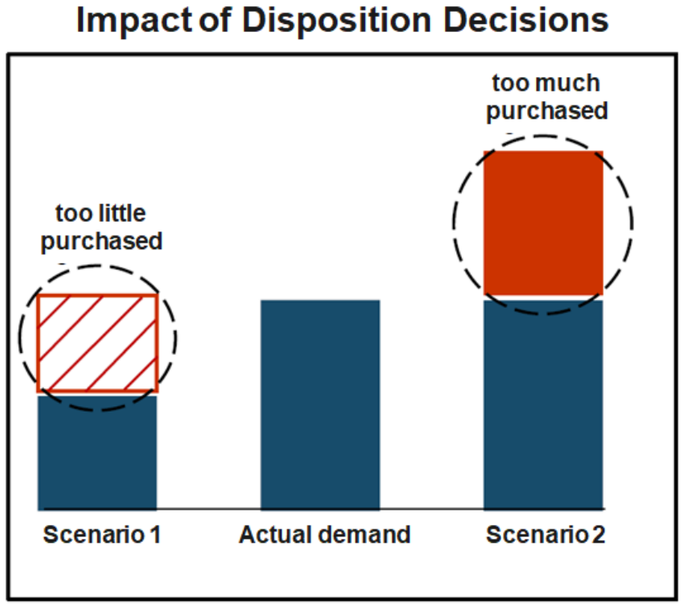
Descriptive Analytics Process



Predictive Analytics Process



Example Inventory Planning



Scenario 1: too little purchased

- Loss of turnover
- Increased procurement costs
- Customer annoyance
- Increased contact costs call center

Scenario 2: too much purchased

- Increased storage costs
- Capital lockup
- Utilization losses

On the basis of more than 300 million data records per week, Otto makes more than one billion forecasts per year on the sales development of individual articles for the next days and weeks. Such forecasts allow Otto to reduce its own inventories by up to 30% on average.

Source: http://www.bvl.de/misc/filePush.php?mimeType=application/pdf&fullPath=/files/441/442/777/1015/DLK12_C3-3_Praesentation_Stueben.pdf

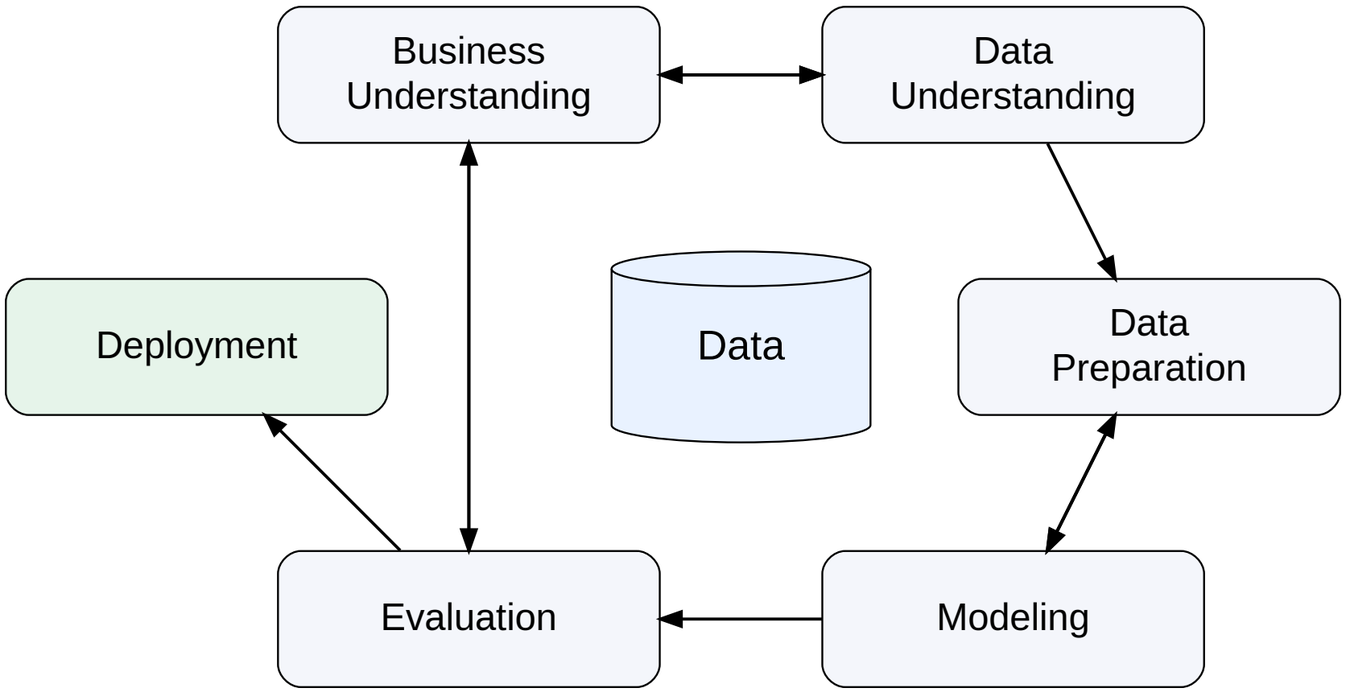
Analytical models

Discipline	Model Culture	Typical Models
Statistics	Probabilistic inference	Regression, GLM, Bayesian
Econometrics	Causal modeling	IV, panel models
Computer Science	Algorithmic learning	Trees, SVM, NN
Operations Research	Optimization	LP, MIP
Management Science	Decision modeling	Stochastic optimization
Complex Systems	Simulation	ABM, DES

-> highlight what we focus on. Mention that not all disciplines focus on the “business understanding” and using insights to support business decisions. Oftentimes, the goal may also be to understand and explain, not to decide (prescribe).

TODO: replace abbreviations

CRISP-DM


Business Understanding
Data Understanding
Deployment
Data
Data Preparation
Evaluation
Modeling

Summary

🎯 **Learning Objective**

- **Illustrate** how modern data analytics capabilities are enabled by data availability, advances in computing power, new algorithms, and modern analytics processes.
- **Distinguish** between descriptive, predictive, and prescriptive analytics.
- **Explore** the Python and Jupyter analytics ecosystem (*exercise*).

Exercise

TODO

- Github account, start Codespace in analytics-and-big-data-notebooks
- option to use a local setup after this session (if it works on your machine)
- explain setup
- Working example (setup time seems to be fast!): https://github.com/github/codespaces-jupyter
- Github advice: https://docs.github.com/en/codespaces/developing-in-a-codespace/getting-started-with-github-codespaces-for-machine-learning